

# MITRAYAN MALLICK

## AI Engineer | Full-Stack Developer

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## ABOUT ME

B.Tech graduate in Computer Science (AI/ML), CGPA **8.79/10**, with hands-on experience building and shipping AI-integrated web applications and Android apps.

Designed and deployed a production Chrome extension with multi-LLM support (Groq, OpenAI, Anthropic) and built a full-stack RAG system with retrieval-augmented generation using LangChain and FastAPI.

Comfortable across the full Software Development Lifecycle (SDLC) — from system design and REST API development to frontend delivery and production deployment on Vercel and Render.

Looking to contribute to a product engineering team as an **AI Engineer or Full-Stack Developer**, where strong fundamentals meet practical AI integration.

## KEY ACHIEVEMENTS

- Shipped **Glimpse**, a production Chrome extension integrating 4 LLM providers (Groq, OpenAI, Anthropic, OpenRouter) with real-time token streaming. Achieved consistent **sub-2-second response latency** with zero-backend architecture, eliminating server cost and latency overhead entirely.
- Built **SecondBrain**, a full-stack RAG document assistant using LangChain, Groq, and MERN. Implemented source-backed citation retrieval that demonstrably **reduced irrelevant AI-generated answers** in internal testing across multi-document queries.
- Engineered **MediCure's** offline-first reminder system at Webguru Infosystems — solving persistent notification failure across low-connectivity devices — resulting in a measured **improvement in on-time medication intake** across 50+ test users during the internship evaluation period.
- Identified **3 critical exploits** on a 10,000+ user gaming platform at Raitomira/Nexon. Filed structured reproduction reports that led to engineering patches within 48 hours, **preventing large-scale abuse** during high-traffic sessions.

## TECHNICAL SKILLS

**Programming Languages:** Python · JavaScript (ES6+) · TypeScript · Java · C · C++ · SQL

**AI/ML & LLM Engineering:** LangChain · RAG · Prompt Engineering · Groq · OpenAI API · Anthropic API · Stable Diffusion · Scikit-Learn · Machine Learning · Deep Learning

**Frontend Development:** React 18 · Next.js · Vite · Tailwind CSS · Framer Motion · Chrome MV3 · HTML5 · CSS3

**Backend Development:** Node.js · Express · FastAPI · REST APIs · Object-Oriented Programming (OOP) · SDLC

**Databases:** MongoDB · PostgreSQL · MySQL · Redis · SQLite · Room DB

**Cloud & DevOps:** Git · GitHub Actions · CI/CD · Docker · Vercel · Render · Linux · AWS (familiar) · Azure (familiar)

**Security:** Vulnerability Analysis · Threat Detection · Exploit Documentation · Incident Reporting

**Tools & Platforms:** VS Code · Android Studio · Postman · GitHub · Figma · Vercel · Render

## EXPERIENCE

### Security & Exploit Analyst · Raitomira / Nexon

Aug 2024 – Present · Remote

- Analysed behavioural patterns across a **10,000+ user gaming platform**, identifying and documenting **3 critical exploits** that were patched within 48 hours — preventing mass-scale abuse during peak sessions.
- Developed and maintained an internal threat-detection workflow, consistently flagging **50+ high-risk accounts per month** and reducing exploit-related disruptions over a 10-month period.
- Produced structured vulnerability reproduction reports used directly by the engineering team to prioritise and action security fixes.

### Freelance Full-Stack Developer · Independent Clients

Jan 2023 – Present · Remote

- NES — National Esports Studio:** Designed and delivered a MERN-stack event platform supporting **500+ registrations**, with an admin CMS, dynamic scheduling, and production deployment on Render.
- Dipan Herbals:** Built a full-stack marketing site with headless CMS integration, third-party payment gateway, and an automated order-notification pipeline — reducing manual client operations and improving user retention through a performance-optimised React frontend.

### Android App Development Intern · Webguru Infosystems

Aug 2024 – Oct 2024 · Kolkata

- Resolved a persistent **offline notification failure** in MediCure (a medicine reminder app for elderly users) by architecting a Room DB-backed sync engine with conflict resolution — ensuring reminders persisted across device reboots and poor-connectivity conditions.
- Delivered a real-time doctor compliance dashboard and foreground-service push notification system, contributing to a measurable **improvement in on-time medication intake** across 50+ test users during evaluation.

## PROJECTS

### Glimpse — AI Sidebar Chrome Extension [github.com/MaximuxR93/GlimpseV1](https://github.com/MaximuxR93/GlimpseV1)

React 18 · TypeScript · Vite · Groq API · Chrome MV3 · LLM Streaming

- Designed a zero-backend, privacy-first architecture to avoid server costs and data exposure — all inference happens client-side via direct LLM API calls with user-supplied keys.
- Integrated 4 LLM providers with a unified streaming interface, achieving consistent **sub-2-second response latency** on content pages. Shipped to production as v1.0.0 with real-world usage.

### SecondBrain — RAG Document Assistant [the2nd-brain-28jk.vercel.app](https://the2nd-brain-28jk.vercel.app)

MERN · FastAPI · LangChain · Groq · RAG · PDF Parsing · Vercel

- Implemented a retrieval-augmented generation (RAG) pipeline — chunking, embedding, and indexing uploaded PDFs — so every AI response cites a specific source passage, reducing hallucination in multi-document Q&A.
- Built end-to-end: FastAPI backend for document ingestion and retrieval, React frontend with streaming chat UI, deployed on Vercel with a CI/CD pipeline via GitHub Actions.

### DocScanX — AI ATS Resume Analyser [doc-scan-x.vercel.app](https://doc-scan-x.vercel.app)

Python · FastAPI · Groq · NLP · REST APIs · Vercel

- Built an ATS simulation engine that parses resumes and job descriptions, then returns keyword-match scores, missing skill gaps, and prioritised rewrite suggestions — live in production.

## EDUCATION

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### Bachelor of Technology — Computer Science & Engineering (AI/ML Specialisation)

Brainware University, Kolkata · 2021 – 2025 · CGPA: **8.79 / 10**

## CERTIFICATIONS

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Python with Karel — Stanford University / Code in Place (2023)

The Complete Web Development Bootcamp — Udemy (2022)

Android App Development — Udemy + Webguru Infosystems (2024)